

Euclid 2026 - Problem 7

7.  (a) In a garden, there are roses and carnations, and no other kind of flower. Each flower is either yellow or white. Half of the yellow flowers are roses, $\frac{1}{4}$ of the roses are yellow, and $\frac{1}{4}$ of all of the flowers are white carnations. What fraction of all of the flowers are yellow roses?



- (b) The function f is defined for every real number $x > 0$ in the following way:

$$f(x) = \begin{cases} 0 & \text{if } 0 < x \leq 1 \\ 1 + f(\log_2 x) & \text{if } x > 1 \end{cases}$$

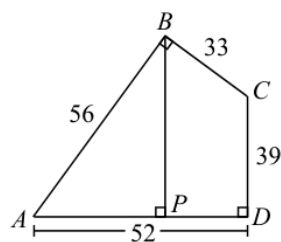
Determine the number of positive integers n for which $f(n) = 4$.

Euclid 2025 - Problem 6

6.  (a) Twelve points are equally spaced around the entire circumference of a circle. In how many ways can three of these points be chosen so that the triangle that they form has at least two sides of equal length?

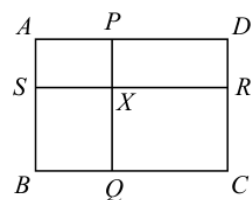


- (b) In the diagram, quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$, $AB = 56$, $BC = 33$, $CD = 39$, and $DA = 52$. Point P is on AD so that BP is perpendicular to AD . Determine the exact length of BP .



Euclid 2024 - Problem 5

5.  (a) In the diagram, rectangle $ABCD$ is divided into four smaller rectangles by the lines PQ and RS , which intersect at X . The areas of these smaller rectangles are, in some order, 2, 6, 3, and a . What are the three possible values of a ?



-  (b) Suppose that the parabola with equation $y = x^2 - 4tx + 5t^2 - 6t$ has two distinct x -intercepts. Determine the value of t for which the distance between these x -intercepts is as large as possible.

Euclid 2023 - Problem 4

4.  (a) Liang and Edmundo paint at different but constant rates. Liang can paint a room in 3 hours if she works alone. Edmundo can paint the same room in 4 hours if he works alone. Liang works alone for 2 hours and then stops. Edmundo finishes painting the room. How many minutes will Edmundo need to finish painting the room?



- (b) On January 1, 2021, an investment had a value of \$400.
From January 1, 2021 to January 1, 2022, the value of the investment increased by $A\%$ from its value on January 1, 2021 for some $A > 0$.
From January 1, 2022 to January 1, 2023, the value of the investment decreased by $A\%$ from its value on January 1, 2022.
On January 1, 2023, the value of the investment was \$391.
Determine all possible values of A .

Euclid 2022 - Problem 3

3.  (a) A large water jug is $\frac{1}{4}$ full of water. After 24 litres of water are added, the jug is $\frac{5}{8}$ full. What is the volume of the jug, in litres?
-  (b) Stephanie starts with a large number of soccer balls. She gives $\frac{2}{5}$ of them to Alphonso and $\frac{6}{11}$ of them to Christine. The number of balls that she is left with is a multiple of 9. What is the smallest number of soccer balls with which Stephanie could have started?
-  (c) Each student in a math club is in either the Junior section or the Senior section. No student is in both sections.
Of the Junior students, 60% are left-handed and 40% are right-handed.
Of the Senior students, 10% are left-handed and 90% are right-handed.
No student in the math club is both left-handed and right-handed.
The total number of left-handed students is equal to the total number of right-handed students in the math club.
Determine the percentage of math club members that are in the Junior section.